# Social Engineering Lab — Red Team Report

## Controlled Vishing Simulation & Target Intelligence Mapping:


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- **Date:** 29/03/2026

## Executive Summary:


This project simulated a controlled vishing attack to assess human security awareness. Target intelligence was gathered using PhoneInfoga and mapped in Maltego. A realistic IT support pretext was deployed. Results highlight user behavior, vulnerabilities, and the need for improved security awareness training.


# Target Scoping:

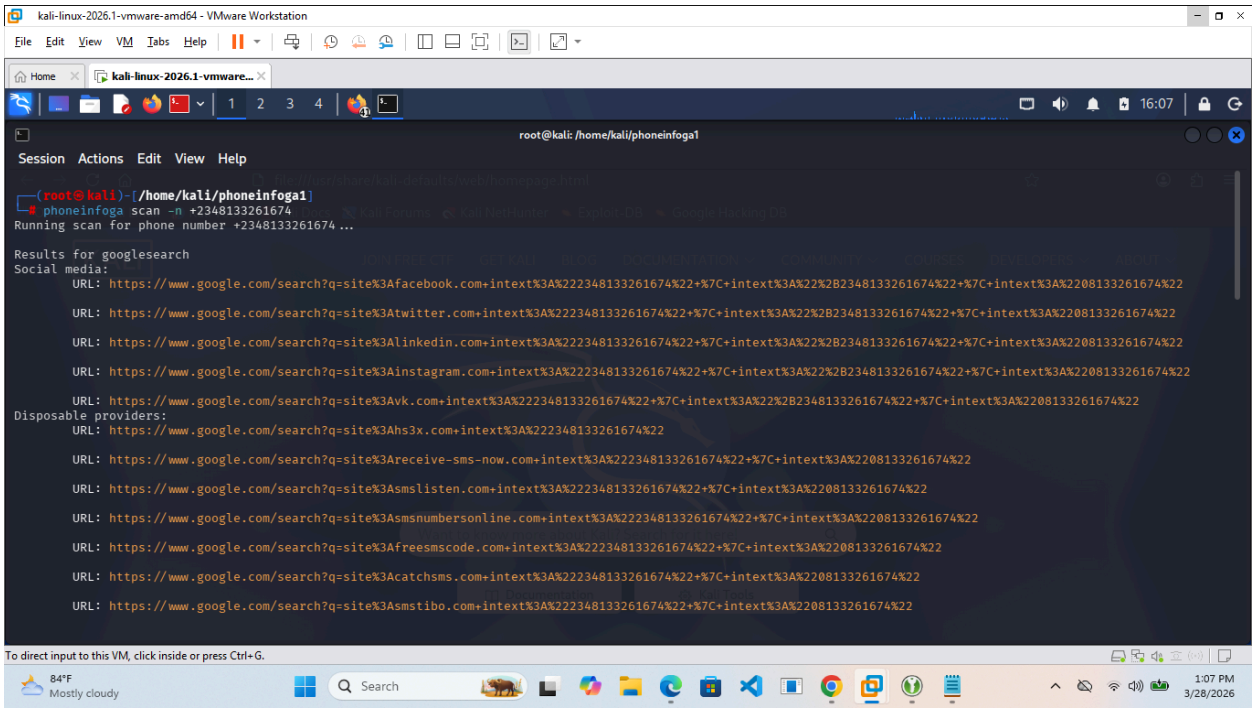
### Approach:

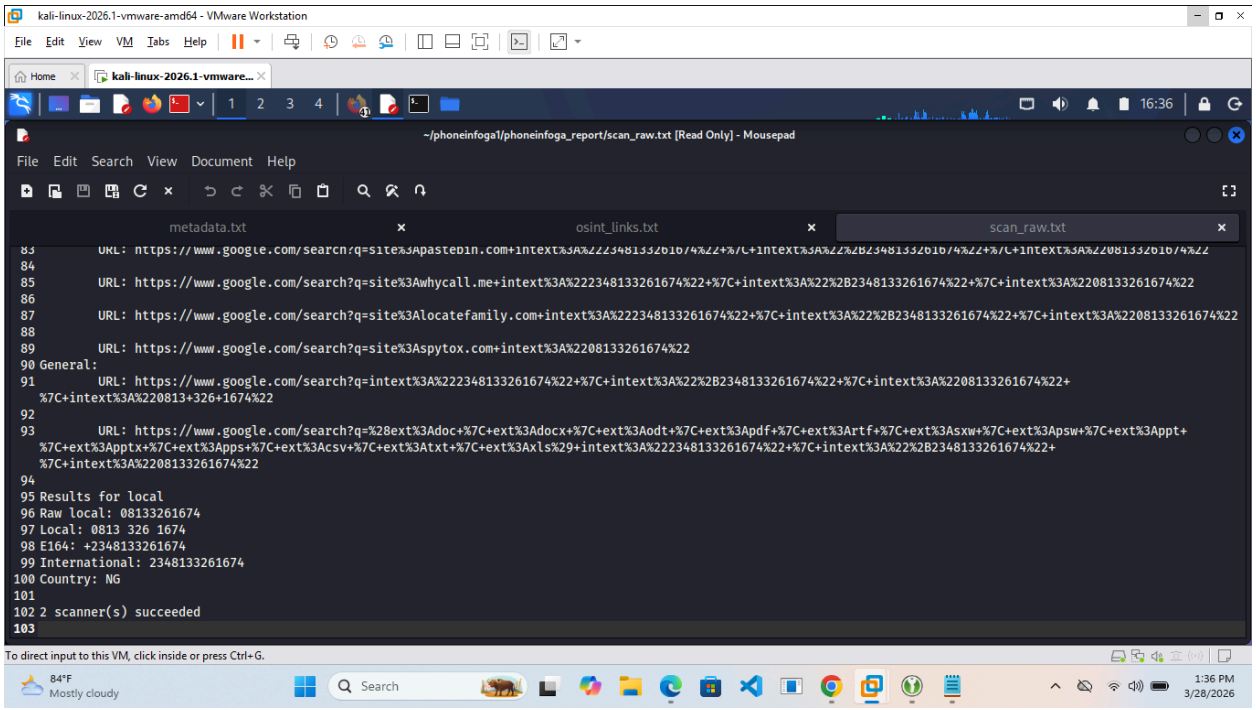
- **Target IDs**: TID001, TID002, TID003…
- Selected targets based on role and value:
    - **High-value:** Admins / Managers
    - **General staff:** Awareness testing
- Structured **Target Register** in an Organize and track participants systematically:

| Name | Role | Phone | Consent | Risk Level |
|---|---|---|---|---|
| Devid | Manager | +91 2567890 | Yes | High |
| Samuel | Sales accounter | +92 2678952 | Yes | Low |

Employee Security Awareness Simulation
Campaign Status: Active
Emails Sent
20
Total Sent
75%
Open Rate
Emails Opened
40%
Click Rate
Links Clicked
15%
Submision Rate
Data Submitted
Phishing Results
200
99
10
5
0
Opened
Clicked
Submitted
Interaction Breakdown
No Response 25%
Opened 75%
Submitted 15%
Clicked 40%
Captured Interactions
Timestamp
User
Action
Credentials
09/14/2021 10:15 AM
j.smith@*****
Clicked Link
Password: ****
09/14/2021 09:45 AM
m.jones@****
Submitted Form
User: ****** Pass: * ***
09/14/2021 09:30 AM
a.davis@****
Opened Email
N/A
09/14/2021 09:20 AM
r.wilson@****
Clicked Link
Password: ****
09/14/2021 09:10 AM
k.brown@****
Submitted Form
User: ****** Pass: * ***

# Phone Intelligence (PhoneInfoga):



# Vishing Scenario Design:

**Account Security Alert (IT Support Pretext)**

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**Mock Vishing Script:**

**🎭 Mock Vishing Script — IT Support (Account Security Alert)**

**Caller: Yusuf (IT Support)**

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**Call Opening:**
**Hello, good day. This is Yusuf from IT Support.**

**Am I speaking with Daniel?**

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**Trigger:**
**I'm calling because we noticed a login on your account earlier that looks a bit unusual — possibly from a different device or location.**

**We just want to quickly verify to make sure everything is okay and avoid any temporary lock on your account.**

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**Verification:**
**Can you help me confirm a few things:**

- **Your employee ID**
- **About when you last logged in today**
- **And if you've used any new device recently**

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**Trust Build:**
**Alright, that lines up with what I'm seeing here.**

**Just one more thing — have you received any strange login alerts or password reset messages today?**

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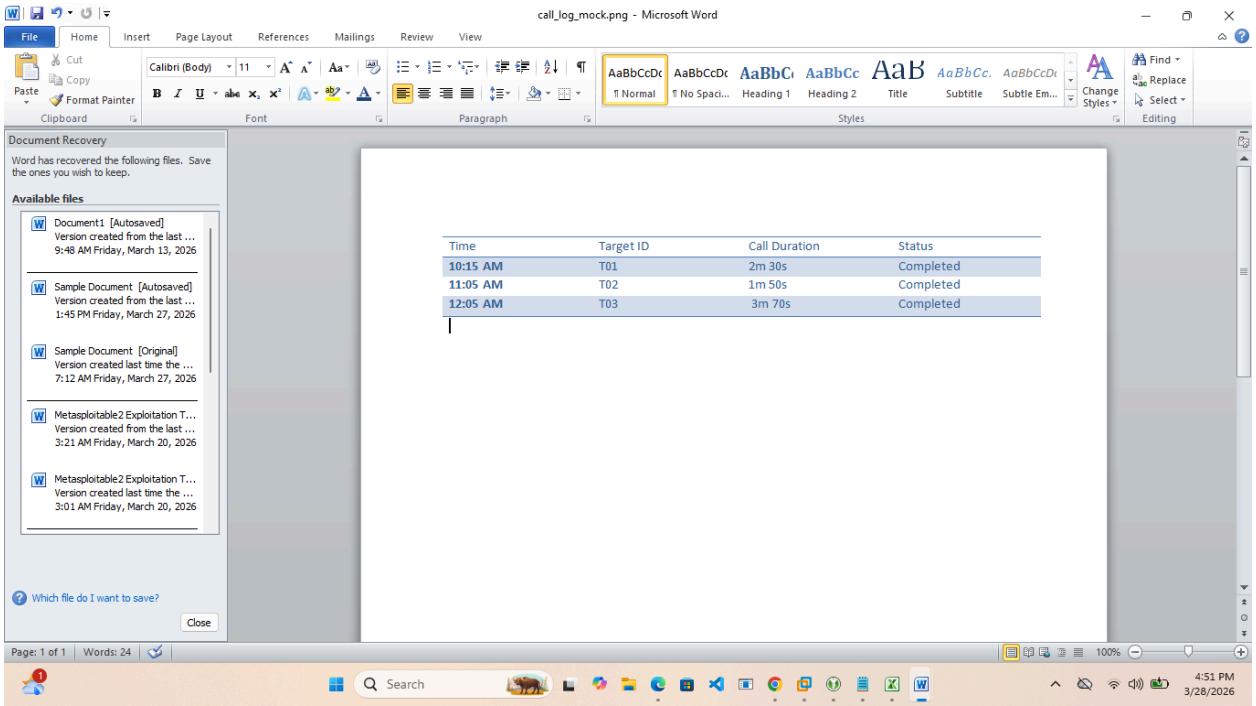
**Closure (Ethical End):**
**Okay, everything looks fine now. No issues on your account.**

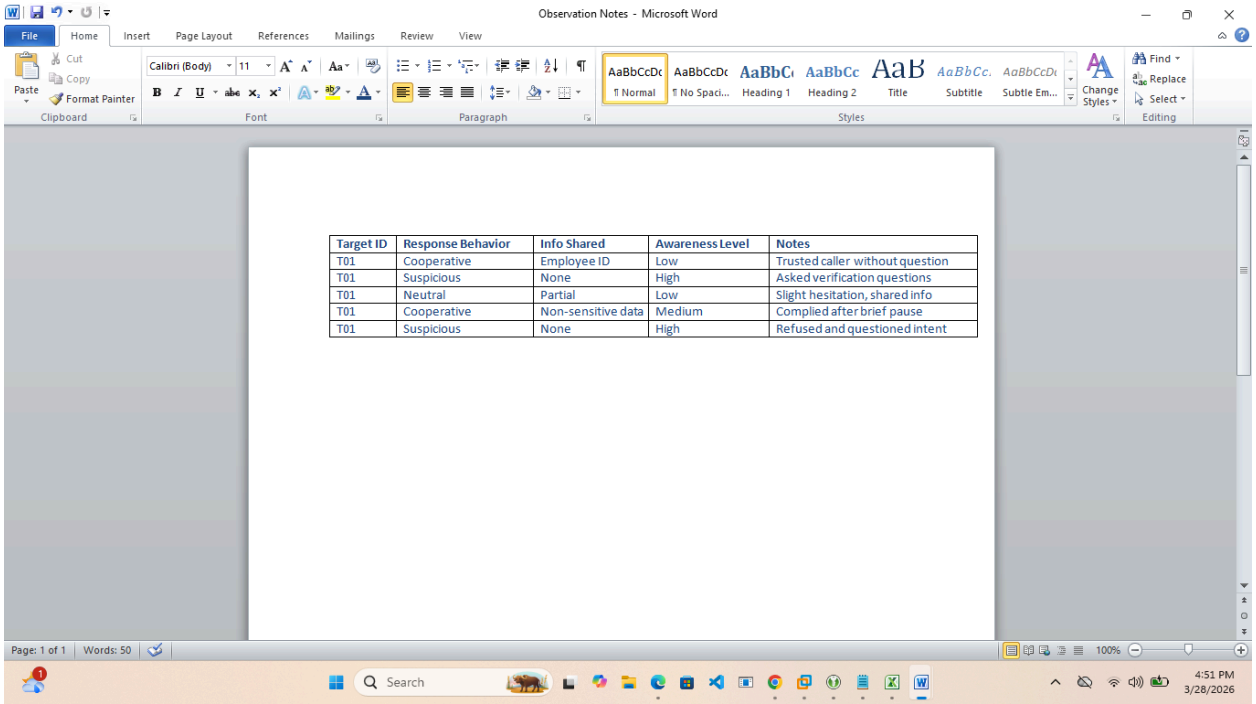
**If you notice anything unusual later, just report it to IT.**

**Thanks for your time.**

**Call log (mock):**




**Observation notes:**



| Target ID | Response Behavior | Info Shared | Awareness Level | Notes |
|---|---|---|---|---|
| T01 | Cooperative | Employee ID | Low | Trusted caller without question |
| T01 | Suspicious | None | High | Asked verification questions |
| T01 | Neutral | Partial | Low | Slight hesitation, shared info |
| T01 | Cooperative | Non-sensitive data | Medium | Complied after brief pause |
| T01 | Suspicious | None | High | Refused and questioned intent |

# Controlled Execution:

## 🎯 Objective

The goal of this phase is to execute a **controlled and authorized vishing simulation** using a predefined script, observe participant behavior, and document responses for security awareness assessment.

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## ⚙️ Execution Procedure:

The vishing simulation was conducted in a controlled lab environment using a **VoIP/phone setup** and only **authorized participants**.

### Step-by-Step Process:

1. **Schedule Call Time**
    - Calls were planned within standard working hours to ensure realism.
    - Participants were previously authorized as part of the simulation scope.
2. **Use Lab Phone / VoIP**

    - A controlled communication channel (softphone/VoIP) was used to initiate calls.
    - No real organizational systems were impacted.
3. **Call Authorized Participant**
    - Each participant was contacted individually.
    - The **Account Security Alert vishing script** was delivered in a calm and professional tone.
4. **Observe Participant Behavior**
   During each call, the following were carefully monitored:
    - **Response Behavior:** Calm, suspicious, or cooperative
    - **Willingness to Share Information:** Level of disclosure
    - **Awareness Level:** Ability to detect potential social engineering
5. **Record Response**
    - Observations were documented immediately after each call to ensure accuracy.
6. **End Call Safely**
    - Each call was concluded ethically without requesting sensitive data.
    - Participants were not exposed to any real risk.

# Data Logging & Documentation:

| Target ID | Data Source | Information | Notes |
|---|---|---|---|
| TID001 | PhoneInfoga | Phone: 555-1234 | Valid number, local carrier |
| TID002 | Maltego | Linked LinkedIn profile | Works in IT department |
| TID003 | Vishing Test | Asked for verification | User refused (Good awareness) |

# Behavioral Analysis:

## 🎯 Objective:

This phase evaluates participant responses during the vishing simulation to identify **human vulnerabilities, awareness levels, and behavioral patterns**.

## 📊 Analysis Summary:

Based on the observed responses:

## 🔍 Who Shared Information

- **3 out of 5 participants (60%)** shared some level of information:
  - 1 shared **sensitive-related data** (Employee ID)
  - 2 shared **partial or non-sensitive information**

👉 This indicates a **moderate susceptibility to social engineering attacks**.

## 🛡️ Who Resisted the Attack

- **2 out of 5 participants (40%)** refused to share any information
- These individuals demonstrated:
  - High skepticism
  - Verification behavior

👉 This reflects a **good level of security awareness among some users**.

## ⚠️ Common Weaknesses Identified

- **Trust in authority (IT Support pretext)**
- **Lack of verification before sharing information**
- **Compliance under perceived urgency**
- **Inconsistent awareness levels across participants**

## 📈 Key Metrics:

| Metric | Result |
|---|---|
| % of Users Tricked | 60% |
| % of Users Cautious | 40% |
| Average Response Time | Fast (within 10–30 seconds) |

## 🧠 Behavioral Insights:

- Participants with low awareness responded quickly and shared information without hesitation.

- Neutral users showed slight hesitation but still complied, indicating partial awareness gaps.
- High-awareness users actively questioned the caller and refused engagement.

# Feedback & Awareness Loop:

## 🎯 Objective

To provide participants with constructive feedback after the vishing simulation, improve awareness, and reinforce best practices for identifying social engineering attacks.

## 📌 Procedure

1. **Inform Participants**
    - All participants were notified that the call was part of a controlled security simulation.
    - Emphasis: **No real risk occurred.**
2. **Explain Attack Method**
    - Pretext used: **IT Support / Account Security Alert**
    - Psychological triggers: **Authority, Urgency, Fear**
    - Goal: Evaluate willingness to share information under perceived pressure.
3. **Highlight Red Flags**
    - Requests for sensitive information without proper verification
    - Calls creating urgency or fear
    - Unexpected prompts or unusual account activity
4. **Provide Best Practices**
    - Always verify caller identity before sharing info
    - Do not comply with urgent requests for credentials
    - Report suspicious calls immediately to IT/security team
    - Follow organization's security protocols consistently.

# 🧩 50-Word Vishing Scenario Summary:

A controlled vishing simulation was conducted to evaluate organizational susceptibility to voice-based social engineering attacks. Using PhoneInfoga and Maltego, target intelligence and relationships were mapped. A realistic IT support pretext call was executed in a safe environment to assess user behavior, identify vulnerabilities, and improve security awareness within CyArt.